

Test of Hypothesis

Hypothesis: Any Verifiable Statement is called a hypothesis.
ie., A Statement, that can be verified whether it is true (or) false,
(or) Accept (or) Reject, through experimentation is known as hypothesis.

Statistical hypothesis:- Any statement about the population from which the samples are drawn. A statistical hypothesis is some assumptions (or) statement, which may or may not be true, about a population, which we want to test on the basis of the evidence from a random sample data.

If hypothesis completely specify the population, then it is known as simple hypothesis, otherwise it is known as composite hypothesis.

In sampling from a normal population with mean (μ), & Variance (σ^2) then hypothesis $H: \mu = \mu_0$ and $H: \sigma^2 = \sigma_0^2$, (where μ_0 is the specified mean of the population and σ_0^2 is the specified variance of the population) are called simple hypothesis. Since it is completely specified the distribution on the other hand each of the following hypothesis is a composite hypothesis:

- (1) $\mu = \mu_0$ but σ^2 is not known
- (2) $\sigma^2 = \sigma_0^2$ but μ is not specified
- (3) $\mu < \mu_0$, $\sigma^2 = \sigma_0^2$ (4) $\mu > \mu_0$ & $\sigma^2 = \sigma_0^2$ etc...

Null hypothesis:- Null hypothesis is "a definite statement about the population parameter." Such a hypothesis, which is usually a hypothesis of no difference, is called null hypothesis. It is denoted by H_0 .

According to Prof. R. A. Fisher, null hypothesis is the hypothesis which is tested for possible rejection under the assumption that it is true.

Ex: (1) $H_0: \mu = \mu_0$; (2) $H_0: \mu_1 = \mu_2$ (3) $H_0: \sigma_1^2 = \sigma_2^2$

Alternative hypothesis :- Any hypothesis which is Complementary to the null hypothesis is called alternative hypothesis.

It is denoted by H_1 , for example, if we want to test the null hypothesis that the population has a specified mean (μ_0) i.e., $H_0: \mu = \mu_0$ then (1) $H_1: \mu \neq \mu_0$ (i.e., $\mu > \mu_0$ or $\mu < \mu_0$)

(2) $H_1: \mu > \mu_0$

(3) $H_1: \mu < \mu_0$

The alternative hypothesis in (1) is known as a two-tailed alternative and the alternatives in (2) and (3) are known as right-tailed and left-tailed alternative respectively.

Errors in Sampling :- The main objective in sampling theory is to draw valid inferences about the population parameters on the basis of the sample results. In ~~such~~ practice we decide to accept or reject the lot after examining a sample from it.

Type I Error : "Reject H_0 when it is true."

Probability of rejecting H_0 when it is true is known as the size of the type one error and it is denoted by ' α '

i.e. $\alpha = P(\text{reject } H_0 \mid \text{When } H_0 \text{ is true})$ is also called

Producer's risk. [i.e., $P(\text{Reject a lot when it is good}) = \alpha$]

Type II Error :- "Accepting H_0 when it is false"

Probability of Accepting H_0 when it is not true is known as the size of the type II error and it is denoted by ' β '

$\beta = P(\text{Accept } H_0 \mid H_0 \text{ is not true})$ is also called Consumer's risk
[i.e., $\beta = P(\text{Accept a lot when it is bad})$]

Level of Significance :- Probability of type-I error is called level of significance, it is denoted by α (alpha)

In any testing procedure it is necessary to fix up the level of significance otherwise not possible to take decisions.

Critical region :- From a population it is possible to draw several samples. The set of all possible samples constitute the sample space. The sample space is divided into two regions namely acceptance region and rejection region.

"The region of rejection of the null hypothesis is called critical region" i.e., If a sample point falls in the critical region then the null hypothesis is rejected. The critical region is denoted by α and the acceptance region is denoted by $1-\alpha$.

"Level of significance is the size of the type-1 error"

One-tailed and Two-tailed tests :- In any test, the critical region is represented by a portion of the area under the probability curve of the sampling distribution of the test statistic.

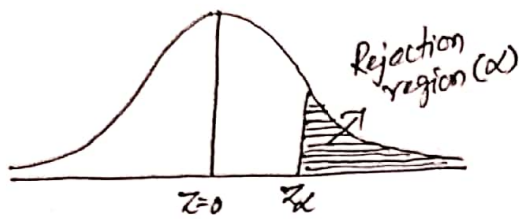
A test of any statistical hypothesis where the alternative hypothesis is one-tailed (right-tailed (or) left-tailed) is called a one-tailed test.

Ex:- A test for testing the mean of a population $H_0: \mu = \mu_0$ against the alternative hypothesis:

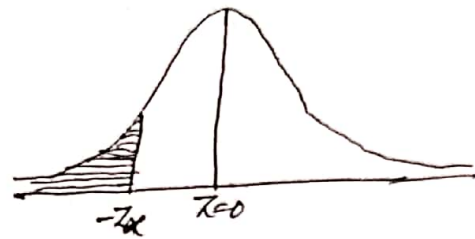
$H_1: \mu > \mu_0$ (Right-tailed) (or) $H_1: \mu < \mu_0$ (Left-tailed)

is a single-tailed test.

Right-tailed test
(Level of Significance ' α ')



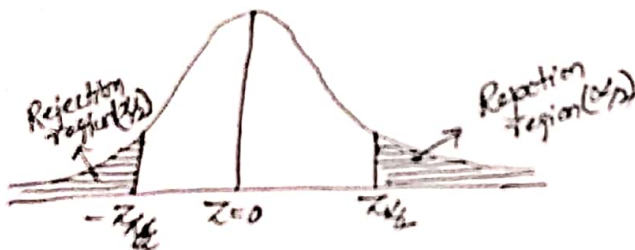
Left-tailed test
(Level of Significance ' α ')



Two tailed test :-

A test of hypothesis where the alternative hypothesis is two tailed Such as : $H_0: \mu = \mu_0$, against the alternative hypothesis $H_1: \mu \neq \mu_0$ ($\mu > \mu_0$ and $\mu < \mu_0$), is known as two-tailed test, and in such a case the critical region is given by the portion of the area lying in both tails of the probability curve of the test statistic.

Two tailed test
(Level of Significance α)



Critical values (or) Significant values : The value of the test statistic which separates the Critical (or rejection) region and the acceptance region is called the Critical value (or) Significant value

Critical value (z_α)	Level of Significance (α)		
	1%	5%	10%
Two-tailed test	$ z_\alpha = 2.58$	$ z_\alpha = 1.96$	$ z_\alpha = 1.645$
Right-tailed test	$z_\alpha = 2.33$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Left-tailed test	$z_\alpha = -2.33$	$z_\alpha = -1.645$	$z_\alpha = -1.28$

Procedure for testing of hypothesis

We now summarise below the various steps in testing of a statistical hypothesis in a systematic manner.

Step-1: Set up the null hypothesis. H_0

Step-2: Set up the alternative hypothesis. H_1 .

This will enable us to decide whether we have to use a single-tailed (right (or) left) test (or) two-tailed test.

Step-3: Choose the appropriate level of significance (α) depending on the reliability of the estimates and permissible risk. This is to be decided before sample is drawn, i.e., α is fixed in advance.

Step-4: Test statistic (or) Test criterion),

$$Z = \frac{t - E(t)}{S.E(t)}, \text{ under } H_0$$

Step-5: Conclusion: We compare the computed value of Z in step-4 with the significant value (tabulated value) Z_α at the given level of significance; α .

If $Z_{cal} < Z_{tab}$, then we Accept H_0

If $Z_{cal} > Z_{tab}$, then we Reject H_0

Note: (1) Accept H_0 means there is no significance difference between population parameter and specified value.

(2) Reject H_0 means there is a difference significance between population parameter and specified value.